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Earth Observation Data, A Way to Find Suitability for Shelters in the Caribbean Coast of Nicaragua

Saira Roxana O. Williams^{a*}, Filoteo Gomez Martinez^b

^a Space Generation Advisory Council, Costa Rica, roxysairawilliams@spacegeneration.org

^b National Aeronautics and Space Administration (NASA) Earth Science Application - SERVIR, Science Coordination, United States, filoteo.gomezmartinez@nasa.gov

* Corresponding Author

Abstract

According to the National Oceanic Atmospheric Administration (NOAA) many situations can cause floods; some of them are heavy rainfall, ocean waves coming on shores, such as a storm surge, dams or levees breaking, melting snow, and ice. As for Central America, the two main reasons are ocean waves on the shore and heavy rainfall. The Atlantic coast of Central America is subject to destructive tropical storms and hurricanes, especially from July through October, accompanied by high winds and floods.

In 2020, two of the strongest hurricanes on record made landfall on the Atlantic coast of Central America; ETA was the first one, followed by IOTA, which was more potent than ETA. Many indigenous communities living on the Atlantic coast of Central America were also affected by these hurricanes. One of them was the Miskito community of Haulover on the Atlantic coast of Nicaragua, whose coastline area transformed. The mangrove forest was broken, and drinking wells were contaminated by saltwater. A wedge of ocean cut through the middle of the small village. Many years ago, precisely in the case of Nicaragua, indigenous communities found schools and churches as shelters to protect themselves from hurricanes. They are not as strong as they should be to be safe because of the infrastructure or were built in the low-lying area.

During the Free and Open-Source Software for Geospatial (FOSS4G) Hackathon, a solution to determine the safest, most suitable sites for shelters and harvesting local information to assess and identify the best locations to build safe shelters was introduced by the first author of this paper. The result from the challenge yielded some information that allowed us to implement other methods, all in the context of hurricane shelters in the Caribbean region in Nicaragua, so that residents can be safe during floods and winds often associated with hurricanes.

This paper will cover the tool concept, design, and solution provided during the Hackathon to tackle this big issue that the community faces yearly. Additionally, the solution links perfectly with Goal #2 of the UN Climate Change Conference (COP), which states, “Adapt to protect communities and natural habitats.” As the COP Goal #2 explains, climate change is already happening, and natural hazards such as hurricanes, rainfalls, and floods are difficult to prevent. However, we can find innovative and creative solutions to protect the community and restore the ecosystem.

Keywords: Earth Observation, Indigenous, Miskito, Hackathon, Shelters, Hurricanes

Acronyms/Abbreviations:

SGAC - Space Generation Advisory Council

FOSS4G - Free and Open-Source Software for Geospatial

NOAA - National Oceanic Atmospheric Administration

CA - Central America

EO - Earth Observations

GEE - Google Earth Engine

FAO - The Food and Agriculture Organization

SEPAL - System for Earth Observation Data Access, Processing and Analysis for Land Monitoring

TEK - Traditional Ecological Knowledge

UN - United Nations

INPE - Instituto Nacional de Pesquisas Espaciais

SST - Sea Surface Temperature

1. Introduction

In the last decades, climate change can be more visible through natural hazards, and these are due to the influence that human civilization has had on the Earth System. Earth Observation data (EO) is a valuable tool that can be used to assess, mitigate, and take actions to create resilience for the communities that are affected the most. EO data use is important, allows decisions makers and leaders of the space sector and the governments agreed and recognised how critically important is “transparent and accountable scaling-up of appropriate public-private cooperation to exploit the contribution to be made by a wide range of data, including EO and geospatial information, while ensuring national ownership in supporting and tracking progress”.

This paper showcases how EO data can be used to find suitability for shelter sites and create resilience on Nicaragua's Caribbean coast during floods due to hurricanes or heavy rainfall. The Caribbean coast of Nicaragua is one of the poorest regions in the country and lacks the infrastructure to support disasters caused by natural hazards. Throughout this paper, we will cover valuable information on creating resilience specifically for the Miskito Indigenous community on the Caribbean Coast of Nicaragua. Therefore, the paper has been divided into six parts:

- In the first section of the paper a brief overview of Nicaragua and climate change is explained.
- In the second part or section of the paper, we cover the history of hurricanes and their impact on Nicaragua and Central America; in this section, we also cover.
- In the third part, the paper provides an overview of Indigenous people, their vulnerabilities, and resilience.
- For the fourth section of the paper, we focus more on using space technologies for Climate change or problem solutions. In this context, we combine how indigenous communities use this data to find solutions to the daily issues they face.
- For section 5, we narrow down all the information we have collected to talk about using EO data in Flood mitigation during hurricanes. For that, we have brought up some case studies or examples of how EO data has been used in other places to mitigate floods.
- In the 6th section, an overview of how Suitability for Shelters App was born, developed initially as a CubeSat Mission design with the aim of testing

technology, creating capacity in Nicaraguan's youth but at the same time providing the images collected by the CubeSat to be used by institutions who are in the frontline during natural hazards.

- A detailed explanation is given in section 7 about how the solution was developed, designed, and maps processed to show the suitability of sites and the areas for shelters. In this section, it is also covered how this project came into the Free and Open-Source Software for Geospatial (FOSS4G) Hackathon organized by the GEO indigenous Alliance.

2. History of hurricanes and its impact in Nicaragua and Central America (CA)

Central American countries are no strangers to hurricanes or tropical cyclones. Hurricane season ranges from July to October; such events are often associated with strong winds and floods, which often cause the destruction of property, crop damage, and loss of lives.

This paper focuses on the Republic of Nicaragua; although centuries of record exist on how hurricanes have impacted Nicaragua, this section provides a list of major hurricanes that have hit Nicaragua in the last 20 years. For example, during Hurricane Mitch in 1998, it is estimated that 3,800 Nicaraguans lost their lives due to flooding. Hurricane Beta in 2005, hurricane Felix in 2007, and most recently Eta and Iota, category 5 and 4 respectively, in November 2020 [3].

2.1 Miskito Community

The Miskito community is an indigenous community located on the north-eastern Caribbean Coast of Nicaragua. Their territory encompasses the western Caribbean region of the country, with an estimated population of 700,000, according to Miskito local authorities. The Miskito community, which is the one we will be covering in this paper, has gone through many social crises, from infrastructure to even having their autonomy violated [4][5].

For decades, Nicaragua has lacked quality education programs. On the Caribbean coast, some children and young people do not even have access to education because they live in remote areas; also, the quality of education is not good. Nevertheless, a couple of ecologists from the community are working hard to learn new innovative technologies and apply their Traditional Ecological Knowledge (TEK) combined with technology to provide solutions to the community. The following link provides an overview of the TEK [6].

One big issue that the Miskito community has faced over the years is that whenever there is a scientific project in the community and for the community, they are not taught how to use the tool or how to analyze the problem. In many cases, Indigenous peoples are not seen as agents of knowledge from the western education system. To remedy such ongoing academic violence from researchers, Indigenous peoples advocate for TEK to be integrated into Indigenous-related studies or in the introduction of new technologies. An example, in the Cultural Survival Quarterly Magazine case of the project “Defending the Miskito Reefs with Maps and GPS: Mapping With Sail, Scuba, and Satellite,” during the process of this project the community at first had no access to the data and imagery that was being used for the project and the initial corporation that was working on the project with funds provided by the USAID said they were training the villagers to use the tools provided even though it was not like that. During the same process of this project, in 1994, the Miskito community found out that the funds provided were used for other areas that did not include the Miskito Reefs [7].

It is vital that the communities are consulted and invited to part from the beginning of a project to completion; doing it enhances research finding, and such research can produce meaningful products and relevance to the Indigenous People. The main reason the project found sites suitable for hurricane shelter sites on the Caribbean coast of Nicaragua is to show the residents the use of EO data and create resilience in the community during hurricanes.

2.2 Haulover village affected by ETA and IOTA

In 2020, two of the strongest hurricanes on record made landfall on the Atlantic coast of Central America; ETA was the first one, followed by IOTA, which was more powerful than ETA. Many indigenous communities living on the Atlantic coast of Central America were also affected by these hurricanes. One of them was the Miskito community of Haulover on the Atlantic coast of Nicaragua, whose coastline transformed, the mangrove forest was broken, and the drinking wells were contaminated by saltwater. A wedge of ocean cut through the middle of the small village, causing destruction to infrastructure and houses and life-threatening to residents.

In section 7 you will be able to see more detailed images on how the two hurricanes affected the community. A big challenge for the whole Caribbean region in Nicaragua is that there are no safe shelters. When there are hurricanes, for example, what the families do is that they use

churches and schools as shelters and none of these places are safe.

In the past the Catholic school Niño Jesus has been used as a shelter even though it is a very old building that is not safe enough to protect the community villagers. In 2020, after ETA and IOTA made landfall in the north region of the Caribbean this school was one of the many schools that were destroyed [<https://www.youtube.com/watch?v=b389QLQgQH8>]



Fig [1]. School destroyed by ETA and IOTA in the Caribbean Coast of Nicaragua

Other places that the villagers have used as a shelter are the churches and the same as the previous school mentioned. These are very old buildings, and they are not built or made for these kinds of natural hazard [<https://www.confidencial.digital/principal/fotorreportaje-haulover-la-comunidad-que-arrasaron-dos-huracanes/>].



Fig [2]. Church destroyed in Haulover due to hurricanes ETA and IOTA

2.3 Causes of Hurricanes

According to the National Oceanic Atmospheric Administration (NOAA) there are many situations that can cause floods, some of which are: Heavy rainfall,

Ocean waves coming on shores, such as a storm surge, melting snow and ice, and Dams or levees breaking. As for Central America, the two main reasons are ocean waves on the shore and heavy rainfall. The Atlantic coast of Central America is subject to destructive tropical storms and hurricanes due to its location, especially between the months of July and October. High winds and floods accompany these hurricanes.

Additionally, as Earth's lower atmosphere is becoming warmer and moisture as a result of human-caused greenhouse gas emissions, this indeed gives the potential for more energy for storms and certain extreme weather events such as hurricanes.

A study was conducted in March 2020 titled "A Century of Warming on Caribbean reefs", some of the things that the study collected was information on Sea Surface Temperature (SST) datasets which calculated the rate of the SST change through time. The study focused in many areas but something impressive found in the combination of analysis in this study was that the Sea surface temperatures are rising across the entire Caribbean basin at a mean rate of 0.04°C per decade since 1871 and more rapidly since 1981 at 0.17°C per decade. In other words, since the SST is increasing this is one of the causes of having hurricanes [26].

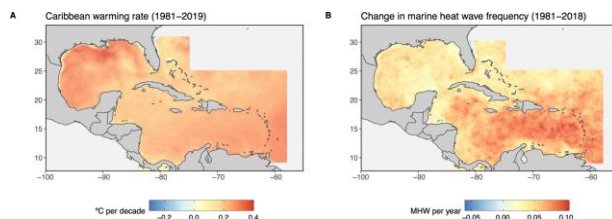


Fig [3]. Temperature rises in the Caribbean Sea

3. Indigenous Peoples, vulnerabilities, and resilience

One of the many reasons Indigenous Peoples distinguish since early contact from other cultures is their adaptability. These adaptations are observed in languages, writing, and reading and recently adopted in developing and using new technologies such as Satellite Imagery. The use and adoption of new technologies are part of the survival strategy. In the following paragraph in this section, we describe some of those examples.

The use of Earth Observation (EO) data provides a critical step for Indigenous self-determination and the preservation of cultural heritage and cultural survival [8].

Similarly, Pierce and Louis also add that Indigenous Communities have successfully used geospatial technologies since the 1970s to document tribal resources

and tribal utility databases and have used them in an informed way [9]. Furthermore, Briggs et al. argue that Geospatial technologies can enable Indigenous data sovereignty [10].

It is important to mention the adoption of new technologies in Indigenous Peoples has been thanks to scholars and non-governmental organizations that channeled on transferring these technologies. Geospatial technologies or EO data should be inclusive, fit within the western concept of universal standards and assimilation, and minimize homogenization to prevent further loss of Indigenous knowledge [11].

Informed engagement of Indigenous communities within the Geospatial Technologies has much to offer to broaden scientific knowledge and include other knowledge to better accommodate the interdependent complexities between the natural environment and human innovation [12].

Furthermore, Chang & Fu, add that Indigenous Peoples are very receptive to learning about satellite data and applications based on their experience with the Atayal Satellites project. Indigenous Peoples of Taiwan increased their interest in studying science; during the project, participants created thematic maps for their tribal villages [13]. Other scholars working with Indigenous peoples suggest that while Indigenous people are eager to learn and embrace new technologies, respect for indigenous cultures, regulatory guidelines, and local customary norms must be implemented [14].

4. Use of Space Technologies for problem solution

This distribution of use of EO data is uneven as opportunities are scarce in many parts of the world. Nevertheless, data derived from spatial technologies are becoming more available even in remote areas. In the United States and Canada, Native/Indigenous peoples have more financial and technical support in contrast to other groups considered as the global south Africa, Asia, and Latin America, where resources and facilities are [15].

Indigenous peoples have generations of knowledge and have so much to contribute and are of great value; if used appropriately, they can contribute significantly to the development of humankind, especially in the complex challenges that climate change impacts and mitigation studies [16].

For example, Indigenous peoples in the Arctic have to define anticipated risks related to rapid change in their local communities concerning climate change and Arctic Industrial development. Reindeer herders empower themselves with new technologies to monitor the changes integrating the best local knowledge possible they have [17].

Indigenous peoples play a significant role in examining and understanding research to understand land use, land cover changes, and other applications. For example, to assess carbon stocks using EO data and in-situ data collected from Indigenous collaborators. Integrating these two data yields a more robust carbon stock estimation. Indigenous Peoples and collaborators who live in the study area play an essential role in the effort to understand local problems and provide meaningful local solutions [18].

4.1 EO data used for tackling climate change issues

In addressing climate variability and change, Indigenous Peoples need to be included and seen as equal partners to better understand what is happening on the ground. Indigenous peoples play a significant role in Guyana in assessing carbon stocks by providing data from the ground combined with satellite imagery; this mesh of data allows for a more robust carbon stock estimation. Moreover, this is a clear example that Indigenous Peoples are needed in order to really understand local problems and solutions [18].

In order to understand local ecological systems and integration of TEK must be integrated into tandem with EO data; by doing this, the resulting output of the projects is more meaningful with high relevance to the Indigenous Communities to implement in their planning and mitigation efforts derived from climate change and variability [19].

5. EO data in Flood mitigation during hurricanes

5.1 Case studies and examples

Integration of EO data must include flood forecasting to mitigate and minimize the negative impact on high-risk communities. Producing flood extent maps can aid humanitarian efforts in the reduction of loss of lives [20].

Not only can EO data assist in emergency response but, more importantly, in the production of flood hazard

zonation and identifications of flood shelter suitability mapping.

One example of flood mitigation is the case of how Uddin & Matin identified suitability areas for building flood shelters concluding that in the low-lying Bangladesh of, the land cover area with 8% extremely suitable, 16% highly suitable, and 7% very low suitable [21]. A similar work exists but it is on assessing residents' flood evacuation shelters, on identifying ideal locations to build flood shelters capturing the community preference is paramount. Flood depth estimation using Sentinel-1(S-1) during and after Hurricane Harvey [22][23].

These examples clearly show the importance of using EO data to tackle challenges vulnerable communities face. As we know due to climate change and its effects, it is almost impossible to stop these natural hazards but what we are capable of doing is to create resilience in order to protect the communities' using technologies.

6. INPE CUBEDESIGN competition

6.1 CubeSat Mission Design - NICASAT

Before getting to the project "Alsut Yawan Tasba Main Kaikai Sika," the initial project for bringing a solution to floods affecting the Caribbean coast was through a CubeSat.

A CubeSat Mission design was developed for technology testing and capacity building. The data collected by this CubeSat was going to be used and processed for finding suitable sites for building shelters. This CubeSat Design was developed during the CUBEDESIGN competition organized by INPE (Instituto Nacional de Pesquisas Espaciais). The competition took around three months. During those three months, a simulation of each subsystem was done using MATLAB and Simulink of the C&DH, EPS, TT&C, ADCS, Communication links, and structures.

The mission of the CubeSat called NICASAT was based on the client's needs: to monitor flood risk areas, deforestation, and fire risk areas in Latin American biomes. From there, the general needs were narrowed down into only one issue and in this case, it was to monitor the Flood risk areas in Nicaragua specifically.

Table 1. NICASAT Mission Goals	
ID	Goals
G1	Tracing high risk flood areas in Nicaragua

- | | |
|----|---|
| G2 | Constantly focused monitoring of the most affected areas of flooding. |
| G3 | Use data collected in order to take actions on the problem. |
-

For the 3 goals established which are mainly general objectives, 2 specific objectives were added under each goal.

Objectives from G1:

1. Cover at least 95% of the national territory.
2. Control the volume of water in real time of big lakes, rivers, and dams.
3. Objectives from G2:
4. Day and night monitoring in the interval when the CubeSat is orbiting.
5. Weekly detection of areas where the water drainage is most harmful to settlements and those where there is higher incidence of flooding.
6. Objectives from G3:
7. Provide data on flood risk areas in Nicaragua that have a probability of accuracy of no less than 98%.
8. Provide real time collected data on the volume of water in rivers, lakes, and lagoons with a veracity of no less than 90%.

The initial mission statement of the CubeSat was just to detect areas of flood risks. After studying the mission deeper, it was adapted to capture images of the Caribbean coast to identify elevation and areas for shelter suitability.

The NICASAT mission completed all of the challenges provided during the competition. It stayed among the top 10 teams who completed all the challenges during the competition. The mission itself is still being worked on and in the future for continuing creating

7. Indigenous Free and Open-Source Software for Geospatial (FOSS4G) Hackathon

The Indigenous FOSS4G Hackathon brings together developers, users, decision-makers, and observers from a broad spectrum of organizations and fields of operation who are indigenous. The competition seeks to have indigenous communities using these tools in order to tackle the challenges that their communities face.

The hackathon helps indigenous people to use their Indigenous heritage, professional and academic background to develop a solution for the challenges provided during the hackathon using space-based technologies, in this case specifically geo data.

7.1 Alsut Yawan Tasba Main Kaikai Sika Challenge

Alsut Yawan Tasba Main Kaikai Sika which means “Let’s protect our lands together” in Miskito a native language from the Caribbean Coast of Nicaragua, was a challenge proposed by Mrs. Saira Williams asked the crowd to develop a tool to help governments and the public in the Caribbean Coast of Nicaragua to determine where shelters for Indigenous Peoples should be built in the event of floods, hurricanes, or storms. Starting from the challenge a solution was produced to determine the safest, most suitable sites for shelters, in addition to harvesting local information to assess site locations and identify the best places to build shelters for the villagers to be safe in during floods specifically caused by hurricanes [25].

7.1.1 Tool concept & design

Scientific literature indicates that floods have been among the top-ranking natural phenomena since 1970. Associated externalities that floods produce are casualties, negative economic impact, etc. EO data can assist flood response operations and evacuation strategies [24].

Sentinel-1 data plays a significant role in flood studies caused by hurricanes, typhoons, cyclones, or tropical torments. Sentinel-1 data is active remote sensing. Its bands penetrate clouds, and water bodies, which is perfect for studying meteorological impacts on the ground, such as floods. In this Alsut Yawan Tasba Main Kaikai Sika Project, Sentinel-1 data was used along with Shuttle Radar Topography Mission (SRTM), road layer from OpenStreetMap (OSM), elevation and land cover data derived from the Food and Agriculture Organization (FAO) SEPAL.



Fig [4]. NICASAT between the Top 10 teams

FAO-SEPAL platform is an open-source project developed by the Open Foris team, which is part of the UN program funded by the government of Norway. FAO-SEPAL tools have transformed the access and use of EO data for easy processing & analysis for land use and land cover monitoring. Using Google Earth Engine (GEE).

Within the GEE, areas that had been flooded before were masked out, then classification thresholds were performed to identify the location to build hurricane shelters using the following thresholds: 1) Low suitability, 2) as suited, and 3) as the best suited to build hurricane shelters. Figure 1 shows the data used and workflow followed.

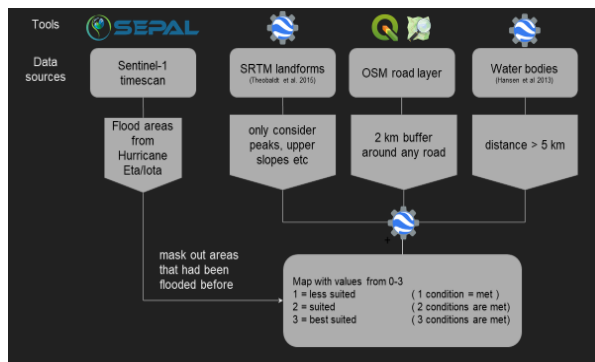


Fig [5]. Data used and required workflow to identify suitability to building hurricane shelters.

7.1.3 Methods used and Solution

The Caribbean coast of Nicaragua of Mosquitia covers around one-third of the country's total land area. This region is a relatively low-lying area, with elevations of 1m to 600m. This low land region is indicated in a black rectangle in Fig 7.

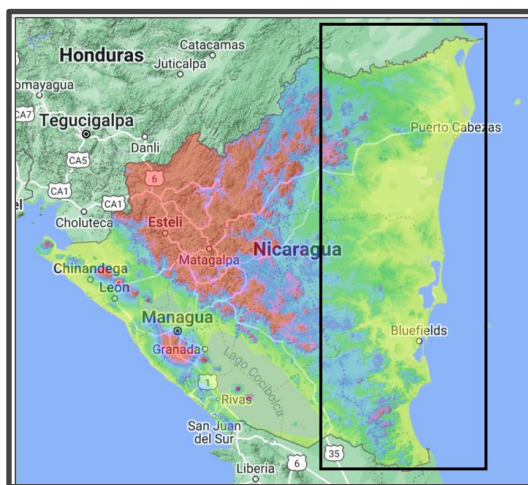


Fig [7]. In black rectangle indicating low region of Nicaragua.

By applying all the criteria described above using the three suitability thresholds to build hurricane shelters, there were very limited options on the ideal location to build hurricane shelters as observed on Figure 3.

There were 3-location identified within the vicinity of the Communities of Puerto Cabezas and Bilwi. And low to medium suitability in Haulover (see, Fig. 5). In color red, it indicates areas suitable for building hurricane shelters.

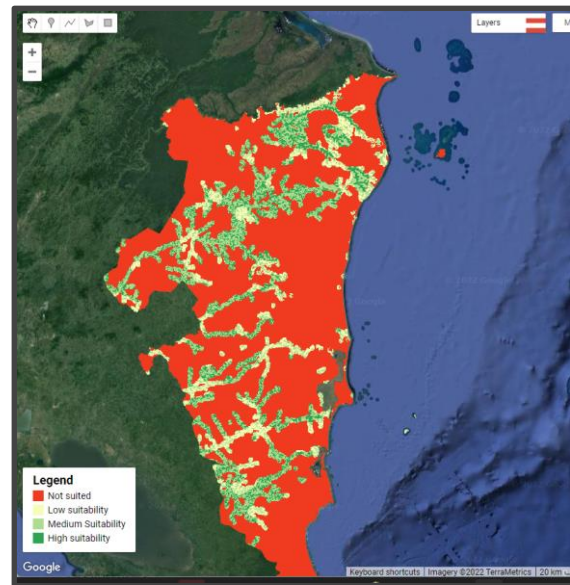


Fig [6]. General overview showcasing suitability threshold on areas where to build hurricane shelters.

The resulting three ideal locations are not surprising as Nicaragua's Atlantic Coast is relatively flat, as observed in Fig. 7. The land cover and vegetation cover are mainly conformed by marsh, swamps, savannas, and mangroves. This low-resulting number of the ideal location to build hurricane shelters relied heavily upon elevation data.

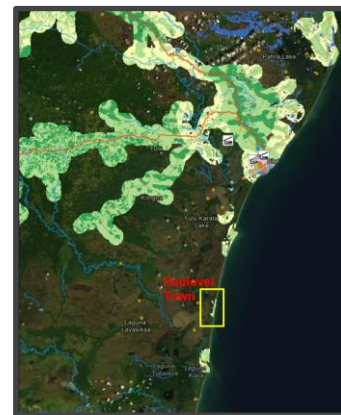


Fig [8]. Haulover town inside of yellow rectangle and surrounding communities.

The Figure 9 is a zoomed-in town of Haulover meets those initial conditions to build hurricane shelter within the community. From low to medium suitability to build hurricane shelters.



Figure 5 Haulover Town zoomed within 100 meters above ground.

7.1.3.1 Image processed

As described earlier in this section, the resulting collaborative work from the Hackathon yielded some results in identifying locations that met the thresholds established in Section 1.1.1 Tool Concept & Design. .

We expanded this work, focusing on Haulover town and other nearby communities. To map the flooded areas during the November 2020 hurricanes, we used Sentinel-1 data from both polarizations VV and VH. Two images were used, the first before the hurricane and the other after Eta and Iota had passed. The resulting product is seen in Figure 6, in blue, indicating flooded areas in Haulover and surrounding communities.

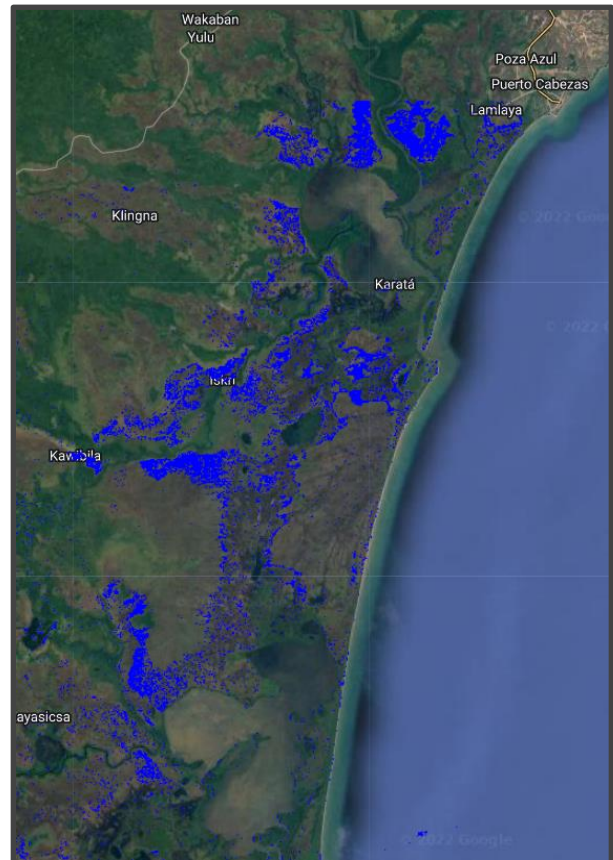


Figure 6. In color blue represent the areas flooded during hurricane Eta and Iota in Haulover and other communities within the vicinity.

Haulover is a coastal community that doesn't often flood during tropical storms compared to other lower-lying areas. We considered it necessary to showcase the flooded areas during the hurricanes Eta and Iota and the associated winds that practically destroyed the whole village. As Observed in Figure 7, in blue color, the ocean water cut across the central town as observed in color blue on the right side of the figure. While such a flood map provides some information, it does not showcase the devastating damages those hurricanes caused in Haulover.

It is not surprising that the flooded areas on the map (in color blue) match the aerial images taken during the hurricanes reported by New York Times and other media outlets.

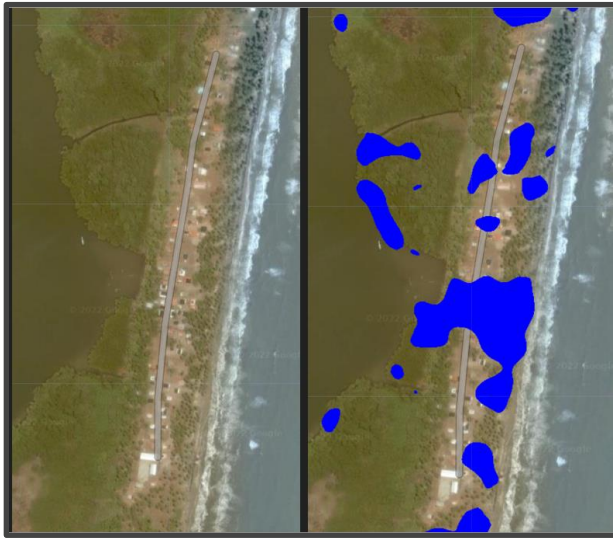


Figure 7. Image on the left is Haulover Community before the hurricane and image on the right the color blue indicating the flooded area during the hurricanes Eta and Iota.

We also wanted to showcase the devastating impact of Landsat 8 data, using to calculate Normalized Difference Vegetation Index, Normalized Difference Water Index, and Normalized Difference Built-up Index; unfortunately, we were unable to find images that had less than 20% of cloud cover of Haulover. As we consider it crucial to understand the damages, we will explore other data with higher resolution, such as Planet monthly composite data, in an effort to understand the damage and assess how the community is recovering.

This is an ongoing work, and the authors will continue to strive to create further image processes and analyses to understand Eta and Iota's aftermath better. Our ultimate goal in this project is the completion of building hurricane shelters for the Haulover community and other vulnerable Atlantic coastal communities in the effort to reduce vulnerability and build resilience for the residents.

7. Conclusion and Recommendations

EO data is an important tool for many challenges and there are many examples we have been able to see through this manuscript. Knowledge, diversity, and inclusion are key for providing solutions to a problem. Having diverse teams makes solutions more inclusive and more creative. Indigenous communities have had the ability to combine their traditional knowledge with technology and provide their perspectives on specific issues which is the main reason this project was done, amplifying the voices of vulnerable communities, giving support and knowledge to the community.

Some of the recommendations we would like to give in order for you to also apply this in your community, understand the images and also share the issues your community faces.

1. Make sure to have people from the community being part of your team since they are the ones who better understand the situation.
2. Search as many resources as you can, but always remember your main resource is the local person you have in your team.
3. While analyzing the images and processing them also compare them with actual pictures of the area.
4. Compare both elevation but also no elevation lands in order to provide suitability sites for both areas.

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